

6 A Apply COM to the collision:

$$(5.0)(200) = (5.0 + 95) u$$

$$u = 10 \text{ m s}^{-1}$$

Apply COE after collision:

$$\frac{1}{2} (100) (10^2) = (100)(9.81) H$$

$$H = 5.1 \text{ m}$$

7 C The 3 lines of action of the forces should coincide at a point